SST MARKER SPACE REVIEW REPORT

Team name: Team Tesla

Team ID: SSTM26_020

Review Number: 3

Submission Date: 30th, June 2026

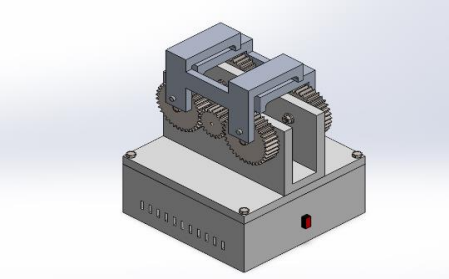
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1. Project Overview & Progress Summary

1.1 Objective of the Bot The primary objective of this robot is to function as a specialized rope-climbing machine capable of traversing a suspended line. To achieve this, the design utilizes a central H-bracket engineered to securely grip the rope. This bracket is driven by a dual-sided gear train mounted on a stabilized base chassis, which translates the motor's power into consistent and controlled movement along the rope's axis.



1.2 Progress Made Since Last Review

- Completed the comprehensive 3D CAD modeling of the primary drive mechanism, structural supports, and chassis.
- Finalized the dual-gear train design and ratios to ensure sufficient torque for lifting the machine's weight during traversal.
- Successfully integrated the top H-bracket gripping mechanism with the primary drive gears.
- Designed the base housing to accommodate the motors, power supply, and control circuitry, incorporating ventilation slots for adequate heat dissipation.

1.3 Key Focus for the Next Phase

- Transitioning from digital modeling to the physical fabrication and 3D printing of the structural chassis and gear components.
- Installing the electrical components, including the motors and internal wiring, into the base housing.
- Conducting empirical friction and payload testing on the H-bracket using standard rope materials to ensure optimal grip and prevent slippage during traversal.

2. Electrical & Software Development

2.1 Introduction

The electrical and software systems form the core of the Rope Runner robot, enabling controlled movement, image acquisition, distance measurement, and communication between the hardware components. At this stage of development, our focus has been on designing the system architecture, selecting suitable components, testing the motor-drive mechanism, and validating the basic control concepts through simulation before full hardware integration.

The electrical and software teams are working closely with the mechanical team to ensure that the robot's movement, weight distribution, and control system are compatible with the final mechanical design.

2.2 Power System (Battery and Power Distribution)

A stable and reliable power system is essential because the robot operates entirely from batteries while suspended on the rope.

After evaluating the power requirements of the motors, controller, and sensors, a **7.4 V (2S) Lithium-ion battery** was selected as the primary power source because it provides sufficient voltage while remaining lightweight.

The power system is divided into two sections:

- A **high-current supply** for the drive motors.
- A **regulated 5 V/3.3 V supply** for the ESP32 controller, camera module, and sensors.

Separating the motor power from the logic power reduces electrical noise generated by the motors and helps prevent unexpected controller resets during operation.

A buck converter will be used to provide a stable regulated output for the control electronics, while a common ground is maintained throughout the system to ensure proper communication between all components.

Future implementation will also include:

- Power switch
- Voltage regulation
- Current protection
- Power monitoring during testing

2.3 Motors and Motor Drivers

The robot uses DC geared motors to generate the torque required to move the clamping mechanism along the rope.

The motors are controlled using an H-bridge motor driver, allowing the controller to:

- Rotate motors clockwise
- Rotate motors counter-clockwise
- Stop the motors
- Adjust motor speed using Pulse Width Modulation (PWM)

During the current phase of development, motor testing has been carried out to understand the interaction between the motors and the gear system. These tests helped verify that the selected motors provide sufficient torque and operate smoothly under different conditions.

Motor control logic is also being developed to ensure synchronized movement between the left and right drive mechanisms.

Future work will include:

- Speed calibration
- Current monitoring

- Motor synchronization
- Load testing while suspended on the rope

2.4 Sensors (Encoder, Camera and Others)

The robot incorporates several sensors to improve navigation and mission performance.

Wheel Encoder

An encoder will be attached to the drive mechanism to measure the distance travelled along the rope.

The encoder generates pulses during rotation, allowing the controller to calculate distance travelled based on wheel circumference and gear ratio.

This information will be transmitted alongside captured images.

ESP32-CAM

The ESP32-CAM module serves as the robot's primary vision sensor.

It will be mounted facing downward to continuously capture images of the search area beneath the robot.

Captured images will be transmitted wirelessly for victim identification.

Future Integration

Additional sensors may be incorporated depending on testing results to improve navigation accuracy and overall system reliability.

2.5 Wiring and System Integration

Since the robot experiences continuous movement and vibration while travelling along the rope, wiring quality is a critical aspect of the design.

The electrical layout has been planned with the following objectives:

- Short wire routing
- Proper cable management
- Secure soldered connections
- Separation of motor wiring from signal wiring
- Reduced electrical interference

Silicone wires will be used where flexibility is required, while heat-shrink tubing and cable ties will provide mechanical support and strain relief.

As development progresses, the wiring harness will be optimized for both reliability and ease of maintenance.

2.6 Control Logic and Software Progress

The control system is based on the ESP32 microcontroller.

Its responsibilities include:

- Reading sensor data
- Controlling motor direction
- Generating PWM signals for speed control
- Measuring travel distance
- Coordinating image capture
- Communicating with the camera module

Current software development has focused on understanding motor behavior and developing the fundamental motor control logic through simulation before hardware implementation.

Circuit behavior and basic motor control concepts have been explored using **Tinker cad**, allowing the team to validate wiring arrangements and control strategies before physical assembly.

Future software development will include:

- Closed-loop motor control
- Encoder feedback processing
- Automated image capture
- Wireless image transmission
- Integration of motor control with the complete mechanical system

2.7 Simulation and Testing

Simulation has been an important part of the development process.

The electrical team has been using **Tinker cad** to:

- Verify circuit connections
- Test motor control logic
- Understand driver operation
- Validate PWM speed control concepts
- Reduce hardware errors before implementation

Motor and gear testing has also been conducted on the physical components to observe torque transmission, rotational direction, and mechanical response.

These preliminary tests provide a solid foundation for the next phase of hardware integration.

2.8 Future Development

The next stage of development will focus on:

- Completing the full electrical wiring

- Integrating the ESP32 controller
- Installing the battery and voltage regulation system
- Connecting the encoder
- Mounting the ESP32-CAM
- Integrating the software with the mechanical system
- Performing complete system testing under real operating conditions
- Optimizing the robot for stability, efficiency, and reliable performance during the competition

3. Physical Design and Build Progress

3.1 Chassis Structure and Materials

The robot chassis is designed as a narrow open cuboid measuring 14 cm × 16 cm × 4 cm. This frame serves as the primary structural support for all mechanical and electronic components while keeping the overall weight low.

The top surface of the chassis provides a flat mounting platform for the gear assembly, which is secured using bolts and screws to ensure accurate alignment. The drive motor and motor driver are housed within the chassis to protect them from external impact, while other components such as the gear linkages are mounted externally for easy access and maintenance.

During the early prototyping stage, several materials were evaluated. Cardboard was initially selected because it is inexpensive, lightweight, and easy to modify. However, repeated testing revealed that it lacked sufficient rigidity, causing components to loosen or detach during operation. To improve structural stability, a plastic board was later used as the prototype chassis, providing a stronger mounting surface.

For the final prototype, the team intends to use acrylic due to its high stiffness, dimensional accuracy, and durability. Acrylic will provide a more rigid structure, reduce unwanted flexing, and improve the overall reliability of the robot.

3.2 Rope Traversal and Gripping Mechanism

The robot uses a mechanically synchronized rope-climbing mechanism that converts the continuous rotary motion of a DC motor into an alternating crawling motion.

Mechanical Drive

A single DC motor drives two identical gears positioned 180° out of phase. This phase difference ensures that while one side of the mechanism performs the power stroke, the opposite side simultaneously resets for the next climbing cycle.

Each gear is connected to an N-shaped linkage, which converts rotary motion into a reciprocating path. Attached to each linkage is a specially designed hook that engages with the rope. Unlike traditional gripping mechanisms, these hooks do not actively clamp the rope. Instead, they rely on gravity and their geometry to remain securely engaged throughout the climbing process.

The robot incorporates four hooks arranged as two pairs on opposite sides, providing continuous support and improving stability during traversal.

Climbing Cycle

The climbing motion consists of three stages:

- **Step (Power Stroke):** One hook is lowered onto the rope and moves forward as the gear rotates, pulling the chassis upward.
- **Handoff (Double-Support Phase):** At approximately the midpoint of the gear rotation, both hook pairs simultaneously engage the rope. This overlap prevents the robot from losing contact and significantly improves stability.
- **Reset (Return Stroke):** The opposite hook is lifted away from the rope and returns to its starting position, preparing for the next climbing cycle while the other hook continues supporting the robot.

Design Advantages

The chosen mechanism offers several advantages:

- Mechanically simple with minimal electronics.
- Requires only one drive motor for synchronized motion.
- Eliminates the need for sensors or active gripping actuators.
- Provides continuous stability through the double-support phase.
- Produces a smooth and rhythmic crawling motion suitable for rope traversal.
- This design was selected because it offers reliable climbing while reducing the risk of the robot slipping or falling.

3.3 Size and Weight Considerations

The chassis dimensions of 14 cm × 16 cm × 4 cm were selected to provide sufficient space for the electronic and mechanical components while maintaining a compact form factor.

Keeping the robot lightweight has been a major design consideration, as excessive weight increases the load on the climbing mechanism and raises the likelihood of slipping during operation.

The current prototype is still under construction; therefore, the final weight has not yet been determined. The team aims to minimize unnecessary mass while ensuring the chassis remains structurally rigid enough to withstand repeated climbing cycles.

3.4 Stability and Durability

The robot's stability is achieved primarily through its double-support climbing mechanism. During each climbing cycle, at least one pair of hooks remains engaged with the rope, preventing the robot from losing support.

To minimize tipping and rotational movement, the heavier components including the motor and electronics are positioned close to the center of the chassis, keeping the center of gravity aligned with the rope.

Although the current prototype can tolerate slight rope movement, the passive hook design also allows the robot to move naturally with rope oscillations without losing engagement.

The overall reliability of the design is enhanced by its mechanical simplicity. Since the climbing sequence is generated entirely by gears and linkages, the system avoids the complexity associated with multiple sensors and actuators.

The components most susceptible to wear include:

- Hook contact surfaces
- Gear teeth
- Linkage pivot joints

To improve durability, the final prototype will use acrylic structural components, stronger fasteners, and improved hook materials to reduce wear during repeated operation.

3.5 Key Challenges and Planned Improvements

Several engineering challenges were encountered during the construction of the prototype.

Challenge 1: Gear Synchronization

The climbing mechanism requires two identical gears with matching tooth counts and dimensions. Finding compatible gears proved difficult, and mismatched gears resulted in poor synchronization of the climbing cycle.

Improvement:
The team selected matching gears to ensure smooth and synchronized operation.

Challenge 2: Chassis Rigidity

The original cardboard chassis lacked sufficient rigidity. During testing, components frequently shifted position or detached because the material flexed under load.

Improvement
The chassis was replaced with a plastic board, which significantly improved rigidity. The final version will use laser-cut acrylic for even greater structural strength and dimensional accuracy.

Challenge 3: Component Mounting

Repeated movement caused screws and mounted components to loosen over time, affecting the alignment of the climbing mechanism.

Improvement:
Future versions will incorporate stronger fastening methods, improved mounting brackets, and more secure screw connections to maintain alignment during operation.

4. Active Development and Iteration

4.1 Tests Conducted and Results

The development of the Rope Runner followed an iterative prototyping approach, where each test was conducted to validate a specific aspect of the mechanical design before progressing to the next stage.

Gear Rotation Test

The first test focused on evaluating the gear mechanism and ensuring it produced the intended reciprocating motion required to drive the hook linkage.

An initial set of gears was fabricated from cardboard. Although the mechanism demonstrated the desired motion, the gears were too rigid and lacked the precision required for smooth operation, resulting in uneven movement and increased friction.

The cardboard gears were subsequently replaced with plastic gears of identical dimensions. This modification significantly improved the smoothness of rotation, allowing the N-shaped linkage to move consistently throughout the climbing cycle.

Result:

Plastic gears provided smooth, synchronized motion and were selected for continued development.

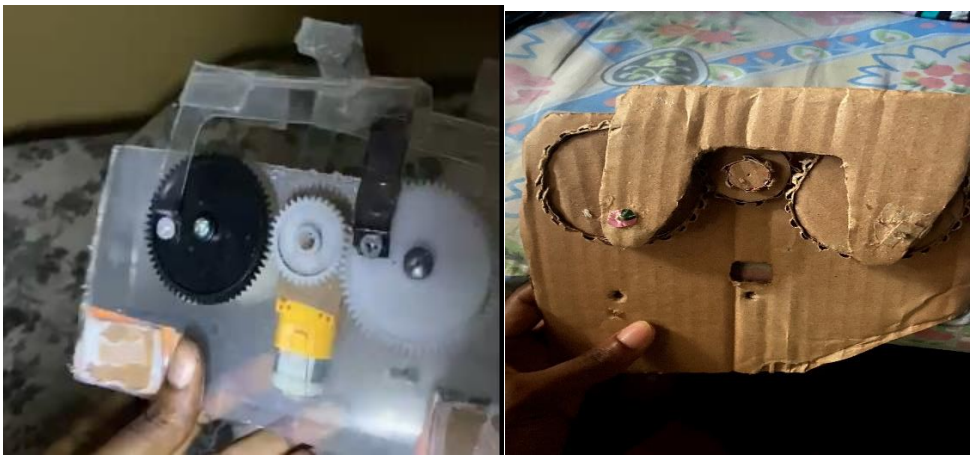


Figure 4.1: Comparison of cardboard and plastic

https://drive.google.com/file/d/1hEyvlmW24g5l4EjwCuTs5M7pu4nQqKpw/view?usp=drivesdk

Video 4.1: Gear rotation test.

Chassis Assembly Test

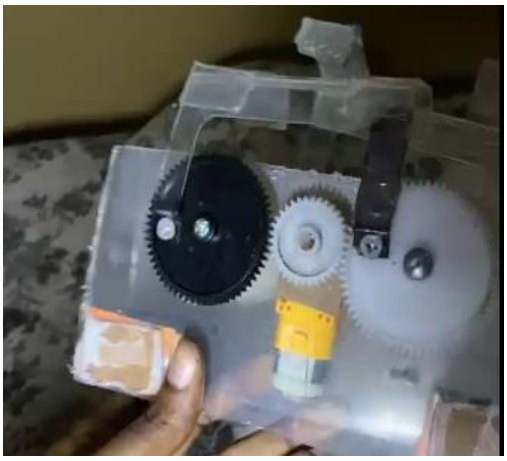
The second test evaluated the structural integrity of the chassis and the mounting of all mechanical components.

The objective was to ensure that the motor, gear assembly, and linkage remained securely mounted during continuous operation.

The initial cardboard chassis lacked sufficient rigidity, causing mounted components to shift or detach during testing. A plastic chassis was then introduced, which improved structural support but still exhibited noticeable flexing under load.

Result

The chassis requires a stronger structural material. Acrylic has been selected for the next prototype due to its superior rigidity and dimensional stability.



Insert Figure 4.2: Cardboard and plastic chassis comparison.

Linkage Geometry Test

Several linkage dimensions were evaluated to determine the geometry required for smooth reciprocating motion.

The linkage measures approximately **15 cm** in length and connects the rotating gears to the climbing hooks. During early testing, incorrect linkage dimensions caused interference between moving components, restricting the mechanism's motion.

Following several design iterations, the linkage dimensions were refined to provide sufficient clearance while maintaining synchronized movement.

Result

The optimized linkage successfully converts rotary motion into smooth alternating movement.

Hook Engagement Test

The hook mechanism was tested to evaluate its ability to remain engaged with the rope during the climbing cycle.

Each hook measures approximately **5 cm × 3 cm**, with a depthTop of the Document of approximately **3 cm**, and is designed to engage ropes with diameters between **10 mm and 15 mm**.

Testing identified two primary issues. First, the hooks occasionally failed to remain fully engaged with the rope because they lacked sufficient rigidity. Secondly, the spacing between adjacent hooks allowed them to become entangled during certain stages of the climbing cycle, temporarily stopping the mechanism.

Result

The hook geometry and spacing require further refinement to improve engagement and eliminate interference.

https://drive.google.com/file/d/1Xny67TDwRjkVwtCLzZCZgCqtCM6OUO_2/view?usp=drivesdk

Video 4.2: Hook engagement test.

Motor Performance Test

The robot is powered by a GM DC motor controlled through a motor driver using PWM speed control. Testing confirmed that the motor generated sufficient torque to drive the climbing mechanism.

During early testing with the cardboard chassis, the motor became noticeably warm due to increased friction and mechanical resistance caused by poor alignment and chassis flexing. After replacing the cardboard gears with plastic gears and improving component alignment, the mechanism operated much more smoothly, reducing the mechanical load on the motor.

Result

The motor provides adequate torque for the current prototype and is suitable for continued development.

4.2 Issues Encountered and Solutions Applied

Issue	Cause	Solution
Rough gear movement	Cardboard gears lacked dimensional accuracy	Replaced with precision plastic gears
Components detached from chassis	Cardboard chassis flexed under load	Switched to a plastic prototype; acrylic selected for the final version
Linkages collided	Incorrect linkage dimensions	Redesigned linkage geometry through multiple iterations
Hooks became entangled	Excessive spacing and unsuitable hook geometry	Planned redesign of hook profile and spacing
Screws loosened during operation	Continuous vibration caused nuts and bolts to rotate loose	Increase mounting-hole clearance where appropriate and use locking fasteners to prevent loosening
Motor heating	High friction caused by poor alignment and chassis flexing	Improved gear alignment and smoother mechanical assembly

4.3 Performance Observations

The robot's performance improved significantly with each prototype iteration.

Replacing the cardboard gears with plastic gears resulted in smooth and consistent motion. The gears meshed accurately, producing synchronized movement of the linkage with minimal vibration.

The climbing mechanism is now correctly synchronized, and the reciprocating motion closely matches the intended design. Although full climbing tests will be conducted after complete assembly, the current mechanical performance indicates that the mechanism functions as expected.

The gear motor has demonstrated sufficient torque to drive the approximately **200 g** prototype without stalling. Motor heating observed during earlier tests was significantly reduced after improving mechanical alignment and reducing friction.

The primary limitation of the current prototype remains the chassis. Although the plastic frame performs better than cardboard, it still flexes under load, affecting long-term stability. Additionally, prolonged operation causes some nuts and bolts to loosen due to vibration.

Overall, the latest prototype represents a substantial improvement over earlier versions and validates the team's mechanical design approach.

4.4 Planned Improvements for the Next Phase

The next stage of development will focus on improving structural rigidity, reliability, and climbing performance.

The team plans to:

- Replace the plastic chassis with laser-cut acrylic to eliminate structural flexing.
- Redesign the hook geometry to improve rope engagement and prevent entanglement.
- Reduce the spacing between adjacent hooks to achieve smoother climbing.
- Modify the mounting holes to allow proper movement of the linkage without causing screws to rotate.
- Replace standard nuts with locking fasteners to prevent loosening during operation.
- Complete the integration of the ESP32, motor driver, potentiometer, and remaining electronic components.
- Conduct full rope-climbing tests on ropes with diameters ranging from **10 mm to 15 mm**.
- Measure climbing speed, stability, power consumption, and endurance during continuous operation.

4.5 Lessons Learned

The iterative prototyping process has provided valuable engineering insights that will guide future development.

The team learned that material selection has a significant impact on mechanical performance. While cardboard was suitable for validating the concept, its flexibility prevented accurate alignment of the gears and structural components. Replacing the gears with plastic demonstrated the importance of manufacturing precision, as it greatly improved synchronization and reduced friction.

The testing process also reinforced that successful rope climbing depends not only on motor power but also on precise mechanical alignment, secure component mounting, and careful linkage design. Small dimensional changes in the linkage and hook geometry had a significant effect on the robot's overall performance.

These lessons will inform the design of the acrylic prototype and contribute to a more reliable and durable climbing robot.

5. Additional Notes

5.1 Bill of Materials / Cost Update

First Purchase

- ESP32-CAM – ₦17,500
- Screws – ₦600
- VAT (7.5%) – ₦45
- Bolts and Nuts – ₦500
- Enclosure – ₦600
- Plastic Gears – ₦5,000
- Transportation – ₦4,000

Subtotal: ₦28,245

Second Purchase

- 2 × Lithium-ion Batteries – ₦1,600
- L298N Motor Driver – ₦2,500
- 2 × DC Gear Motors – ₦1,600
- Power Switches (2) – ₦200
- Battery Holder – ₦1,000
- LM2596 Buck Converter – ₦200
- Connecting Wires – ₦700
- 2 × Servo Motors – ₦2,000
- Battery Charger – ₦1,260
- ESP32 Development Board – Already Available (No additional cost)

Subtotal: ₦11,060

5.2 Component Sourcing Status

Most of the required components for the electrical and mechanical systems have now been successfully sourced. These include the ESP32-CAM, ESP32 development board, DC gear motors, servo motors, L298N motor driver, lithium-ion batteries, battery holder, buck converter, gears, enclosure, switches, wires, fasteners, and battery charger.

With the majority of the components available, the team is now focused on system integration, testing, and refinement of the prototype.

5.3 Team Structure / Role Updates

The team continues to collaborate effectively across the mechanical, electrical, and software divisions.

- **Team Lead:** Coordinates project planning, prototype development, and team activities.
- **Mechanical Team:** Responsible for the climbing mechanism, chassis, gear system, and structural improvements.
- **Electrical Team:** Responsible for power distribution, motor control, wiring, circuit simulation, and component integration.
- **Software Team:** Responsible for motor control algorithms, sensor integration, camera operation, and communication between hardware components.

Regular meetings are held to review progress, identify challenges, and coordinate the next stages of development.

5.4 Risks / Anticipated Delays

During prototype development, several challenges were identified that required modifications to the design:

- The initial cardboard gears lacked the required strength and precision, leading to inconsistent motion. This was resolved by replacing them with plastic gears.

- Early prototypes experienced increased motor heating due to mechanical friction and alignment issues. These were reduced through improved gear alignment and smoother transmission.
- The current plastic chassis still exhibits slight flexing under load, which may affect stability during extended climbing tests.
- Continuous vibration during operation causes some nuts and bolts to loosen over time, requiring additional fastening methods such as lock nuts or thread-locking compound.
- Final climbing tests and complete electrical integration are still pending and may require further calibration before the final competition.

Although these challenges have slightly extended development time, they have also improved the overall reliability and performance of the robot through iterative testing and redesign.

5.5 Support Needed from Makerspace

The team would appreciate continued support from the Makerspace in the following areas:

- Access to 3D printing facilities for manufacturing a stronger and more rigid chassis.
- Technical guidance on optimizing the climbing mechanism and overall system integration.